

Mortgage Pricing Model (MKM-MP-001)

Wolfram Oracle Benchmark of the Fair-Value Calculation

MKM Research Labs

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Contents

1 Executive Summary **3**

2 Model Purpose and Scope **4**

2.1 What is benchmarked 4

2.2 What is not 4

3 Mathematical Framework **4**

4 Benchmark Method **5**

5 Validation and Backtesting **5**

5.1 The double-counted loss 5

5.2 The model document's claims 6

6 Model Limitations and Known Weaknesses **6**

7 Document History **6**

1 Executive Summary

This document benchmarks the Mortgage Pricing Model (MKM-MP-001), which values a flood-exposed residential mortgage from its amortisation schedule, an affordability-based credit spread and a loss given default. It asks whether the code computes what its model document (`mortgage_pricer.pdf`, v1.1) says, and whether what the document says is right. An earlier review suspected that the fair value counts the expected loss twice.

The code computes its documented formulae. Over 360 loans — loan-to-value 0.5 to 1.0, recovery haircuts 0 to 40%, two incomes, two coupons, three terms, discounted at the coupon and at a risk-free rate — the payment, spread, expected cashflows, expected losses and fair value agree with a 50-digit implementation of Eqs. 1–24 to 1.7×10^{-14}. Wolfram and the 50-digit reference agree to 3.8×10^{-30} on both values.

The expected loss is counted twice: confirmed. The document's expected cashflow (Eq. 19) pays the lender, in the month of default, the balance *less* the loss given default. The fair value (Eq. 24) then subtracts the present value of that same loss again. Wolfram values the loan independently, as the expectation over the month of default of the payments made before it and the recovery made in it, and that value equals Eq. 22 alone, with no loss term. The model understates fair value by the full PV of expected loss: on 53% of the loans in the grid, up to 34.8% of par, so a GBP 300,000 loan worth GBP 206,262 is valued at GBP 101,992 (Table 2). The expected-loss outputs are themselves right; the fair value and the discount to par are not. The document's own sensitivity tables never show it, because their baseline (LTV 0.67, haircut 25%) has no loss given default.

Other findings (Table 3):

1. **The recovery is measured a month late.** Eq. 19 pays $B_{i-1} - \mathrm{LGD}_i$ with the loss on the end-of-month balance B_i, so when the loan is under water the lender recovers a month's principal more than the property fetches: up to 0.97% of par.
2. **Payment sensitivity is understated (§6.1).** A 100 bp rise raises a 25-year payment by 8.3–11.9%, not 6–8%.
3. **The zero-income fallback cannot trigger (§3.2.4).** Income is floored at GBP 1 before the spread is computed, so zero income prices at 37.62%, not 15%.
4. **Production does not discount as documented (§3.8).** The loan calculator discounts at a risk-free rate, not the coupon; at the grid's rates that moves fair value by up to 25%, and the discount to par no longer measures credit alone (§5.3).

The document's Table 7 is reproduced exactly, the expected-loss bound of §7.2.2 holds, and Wolfram proves the par identity of §5.3 and §7.2.1 symbolically.

Evidence. 6 checks, outcome: all pass (Table 1).

- Wolfram: 15.0.0 for Mac OS X ARM (64-bit) (May 26, 2026) via live kernel session (wolframclient).
- Code under test, at commit `7bbb63078a59`.

- Numerics: numpy 1.26.4, scipy 1.15.1, mpmath 1.4.1.
- Generated 2026-10-27 by `python -m tools.wolfram_oracle mp001`.

2 Model Purpose and Scope

2.1 What is benchmarked

`LoanPricer.price_loan` in `src/models/loan/pricer/` (`_pricing.py`, with the spread in `credit.py`), imported as production imports it and called as the loan calculator (`src/routes/_loan_pricing`) calls it. The document's §5.1 names `src/models/mortgage_pricer.py` and `MortgagePricer.price_mortgage`, which have since been renamed.

2.2 What is not

The calibration of the spread schedule (§4.3), the coupon the calculator builds upstream of the pricer, and the portfolio aggregation (§3.9), which is arithmetic. This is a benchmark of the implementation (validation question Q8), not independent validation (Q9).

3 Mathematical Framework

Following the document (§3): monthly payment M, balances $B_0, \dots, B_n$, monthly default probability $h = 1 - e^{-s/12}$, survival $S_i = (1-h)^i$, $\mathrm{PD}_i = S_{i-1} - S_i$, recoverable value $R = (1-\delta)V$, $\mathrm{LGD}_i = \max(0, B_i - R)$ and discount factors D_i. Then

$$\mathrm{PV_{CF}} = \sum_i D_i\big[M S_i + (B_{i-1} - \mathrm{LGD}_i)\,\mathrm{PD}_i\big], \quad \mathrm{PV_{EL}} = \sum_i D_i\,\mathrm{LGD}_i\,\mathrm{PD}_i, \quad \mathrm{FV} = \mathrm{PV_{CF}} - \mathrm{PV_{EL}}. \tag{1}$$

The pathwise value. Let τ be the month of default, so that $P(\tau = i) = \mathrm{PD}_i$ and $P(\tau > n) = S_n$. On the path where it defaults in month τ, the lender receives M in each month before τ and the recovery $B_{\tau-1} - \mathrm{LGD}_\tau$ in month τ; on the path where it never defaults, all n payments. Taking the expectation,

$$\mathbb{E}[\mathrm{PV}] = \sum_{i=1}^{n} D_i\big[M\,P(\tau > i) + (B_{i-1} - \mathrm{LGD}_i)\,P(\tau = i)\big] = \mathrm{PV_{CF}}. \tag{2}$$

The loss is already in the recovery. Subtracting $\mathrm{PV_{EL}}$ counts it a second time, so FV understates value by exactly $\mathrm{PV_{EL}}$. Either Eq. 19 should carry the gross balance B_{i-1} and Eq. 24 subtract the loss, or Eq. 19 should stand and Eq. 24 be $\mathrm{FV} = \mathrm{PV_{CF}}$.

The recovery. The property fetches R whatever the balance, so the recovery in month i is $\min(B_{i-1}, R)$. Eq. 19's $B_{i-1} - \max(0, B_i - R)$ equals $R + (B_{i-1} - B_i)$ when $B_i > R$: a month's principal more.

4 Benchmark Method

The equations above are evaluated with `mpmath` at 50 significant digits, from the document rather than the code; the loan-to-value ratio is taken as the double the code compares against its bands, since a band edge such as 0.95 falls between the exact ratio and its double. Wolfram evaluates, from each input's exact value at 60-digit precision, both the document's fair value and the pathwise value, the latter as an expectation over a `GeometricDistribution` default month with the payments' present values accumulated, and it proves symbolically that the contractual payments discounted at the coupon are worth par.

5 Validation and Backtesting

Table 1: Benchmark checks

Check	Result	Tolerance	Outcome
Monthly payment (Eq. 1): model vs document	2.0×10^{-14}	1.0×10^{-12}	pass
Credit spread (Eqs. 6–13): model vs document	6.4×10^{-14}	1.0×10^{-12}	pass
PV of expected cashflows (Eqs. 19, 22): model vs document	1.7×10^{-14}	1.0×10^{-12}	pass
PV of expected losses (Eqs. 20, 23): model vs document	4.0×10^{-14}	1.0×10^{-12}	pass
Fair value (Eq. 24): model vs document	1.7×10^{-14}	1.0×10^{-12}	pass
Wolfram agrees with the reference (fair value and pathwise value)	3.8×10^{-30}	1.0×10^{-12}	pass

5.1 The double-counted loss

Table 2: The document's fair value against the pathwise value: GBP 300k property, 6% coupon, 25 years, discounted at the coupon

LTV	δ	Income	s (%)	Eq. 24	Pathwise	Twice-counted	of par
0.80	0.25	30,000	23.06	230,639	234,880	4,240	1.8%
0.80	0.25	60,000	3.78	237,986	238,826	840	0.3%
0.80	0.40	30,000	23.06	170,304	204,712	34,408	14.3%
0.80	0.40	60,000	3.78	221,043	230,354	9,312	3.9%
0.90	0.25	30,000	34.94	213,773	241,339	27,567	10.2%
0.90	0.25	60,000	5.35	254,743	262,129	7,386	2.7%
0.90	0.40	30,000	34.94	138,483	203,695	65,211	24.2%
0.90	0.40	60,000	5.35	225,724	247,619	21,896	8.1%
0.95	0.25	30,000	48.35	195,529	239,654	44,125	15.5%
0.95	0.25	60,000	7.03	255,737	270,062	14,325	5.0%
0.95	0.40	30,000	48.35	115,840	199,809	83,970	29.5%
0.95	0.40	60,000	7.03	218,248	251,318	33,070	11.6%
1.00	0.25	30,000	59.06	179,117	238,898	59,782	19.9%
1.00	0.25	60,000	9.17	250,966	275,106	24,141	8.0%
1.00	0.40	30,000	59.06	97,635	198,157	100,523	33.5%
1.00	0.40	60,000	9.17	204,922	252,084	47,162	15.7%

5.2 The model document's claims

Table 3: The model document's claims, tested

Claim	Stated	Benchmark finds	Outcome
§1, Eqs. 19, 24	fair value is the PV of expected cashflows net of expected losses	the loss is deducted twice: fair value understated by up to 34.8% of par	**not met**
Eq. 19 recovery	$B_{i-1} - \mathrm{LGD}_i$	exceeds the recoverable value by a month's principal; up to 0.97% of par	finding
§6, Table 7	spread and value by flood category	reproduced to the figures quoted; no loss given default at the baseline	met
§6.1	+100 bp raises the 25-year payment 6–8%	8.3–11.9% over rates of 2–9%	**not met**
§3.2.4	zero income gives a 15% spread	income is floored at GBP 1 first; the spread is 37.62%	**not met**
§7.2.2	$\sum_i \mathrm{EL}_i \le B_0$	at most 36.7% of B_0	met
§5.3, §7.2.1	at the contractual rate the payments are worth par	identity proved symbolically	met
§3.8	discount at the contractual rate	production discounts at the risk-free rate; fair value moves up to 25%	finding

6 Model Limitations and Known Weaknesses

1. **Benchmark, not validation.** The code is shown to compute its documented formulae, and one of those formulae is shown to be wrong; whether an affordability-based spread is the right default model is a separate question.

2. **The fair value.** Set $\mathrm{FV} = \mathrm{PV}_{\mathrm{CF}}$ in `_pricing.py` (and Eq. 24), or make Eq. 19 gross. This raises every fair value with a loss given default, and every downstream figure built on it, so it is for the model owner and the MRC; the expected-loss outputs are unchanged.

3. **The recovery.** Pay $\min(B_{i-1}, R)$ in the month of default.

4. **Documentation.** §6.1 should quote 8–12%; §3.2.4 should be removed or the income floor moved after it; §3.8 and §5.3 should describe risk-free discounting; §5.1 should name the package. §3.9 classifies a loan as high-risk by spread above 10% and §9.5 by expected loss above 5%; §9.9 calls the model Tier 1, the inventory Tier 2.

7 Document History

Date	Version	Description	Author
27-Oct-2026	0.1	First issue	David K Kelly